```
mysql> use company_db;
Database changed
mysql> show tables;
+----------------------+
| Tables_in_company_db |
+----------------------+
| department           |
| employee             |
+----------------------+
2 rows in set (0.00 sec)

mysql> CREATE TABLE orders ( order_id INT PRIMARY KEY, customer_name VARCHAR(100),
order_date DATE, total_amount DECIMAL(10,2), status VARCHAR(20) ) ENGINE=InnoDB;
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE USER IF NOT EXISTS 'john'@'localhost' IDENTIFIED BY 'JohnPass2026!';
Query OK, 0 rows affected, 1 warning (0.05 sec)

mysql> CREATE USER IF NOT EXISTS 'alice'@'localhost' IDENTIFIED BY 'AlicePass2026!';
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE USER IF NOT EXISTS 'intern_user'@'localhost' IDENTIFIED BY
'InternPass2026!';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER IF NOT EXISTS 'temp_user'@'localhost' IDENTIFIED BY
'TempPass2026!';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE USER IF NOT EXISTS 'sec_user'@'localhost' IDENTIFIED BY 'SecPass2026!';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER IF NOT EXISTS 'admin_user'@'%' IDENTIFIED BY 'AdminPass2026!';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER IF NOT EXISTS 'auditor'@'localhost' IDENTIFIED BY 'AuditPass2026!';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT SELECT ON company_db.* TO 'auditor'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE USER IF NOT EXISTS 'analyst'@'localhost' IDENTIFIED BY 'AnalystPass2026!';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT SELECT ON company_db.* TO 'analyst'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER IF NOT EXISTS 'hr_assistant'@'localhost' IDENTIFIED BY 'HrPass2026!';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT, UPDATE ON company_db.employees TO 'hr_assistant'@'localhost';
ERROR 1146 (42S02): Table 'company_db.employees' doesn't exist
mysql> GRANT SELECT, UPDATE ON company_db.employee TO 'hr_assistant'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER IF NOT EXISTS 'finance_user'@'localhost' IDENTIFIED BY 'FinPass2026!';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> GRANT SELECT, UPDATE ON company_db.department TO 'finance_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER IF NOT EXISTS 'team_lead'@'localhost' IDENTIFIED BY 'LeadPass2026!';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT SELECT, INSERT ON company_db.* TO 'team_lead'@'localhost' WITH GRANT
OPTION;
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT SELECT ON company_db.* TO 'john'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT SELECT, INSERT ON company_db.employee TO 'alice'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE INSERT ON company_db.employee FROM 'alice'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT sp_before_update;
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL PRIVILEGES ON *.* TO 'admin_user'@'%' WITH GRANT OPTION;
Query OK, 0 rows affected (0.02 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE ALL PRIVILEGES, GRANT OPTION FROM 'intern_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT DELETE ON company_db.orders TO 'dev_user'@'localhost';
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> GRANT DELETE ON company_db.order TO 'dev_user'@'localhost';
ERROR 1146 (42S02): Table 'company_db.order' doesn't exist
mysql> START TRANSACTION; UPDATE employees SET salary = 80000.00 WHERE emp_id =
101; SAVEPOINT sp_first; UPDATE employees SET salary = 95000.00 WHERE emp_id = 102;
ROLLBACK TO SAVEPOINT sp_first; COMMIT;
Query OK, 0 rows affected (0.00 sec)

ERROR 1146 (42S02): Table 'company_db.employees' doesn't exist
Query OK, 0 rows affected (0.00 sec)

ERROR 1146 (42S02): Table 'company_db.employees' doesn't exist
```

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> REVOKE UPDATE ON company_db.employee FROM 'hr_assistant'@'localhost';

Query OK, 0 rows affected (0.01 sec)

mysql> SET TRANSACTION READ ONLY; START TRANSACTION; SELECT * FROM company_db.employees; SELECT * FROM company_db.departments; COMMIT;

Query OK, 0 rows affected (0.03 sec)

Query OK, 0 rows affected (0.00 sec)

ERROR 1146 (42S02): Table 'company_db.employees' doesn't exist

ERROR 1146 (42S02): Table 'company_db.departments' doesn't exist

Query OK, 0 rows affected (0.00 sec)

mysql> SET TRANSACTION READ ONLY; START TRANSACTION; SELECT * FROM company_db.employee; SELECT * FROM company_db.department; COMMIT;

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

+--------+----------+----------+---------+

| emp_id | emp_name | salary | dept_id |

+--------+----------+----------+---------+

| 1 | Aman | 40000.00 | 10 |

| 2 | Bhavna | 55000.00 | 10 |

| 3 | Chetan | 30000.00 | 20 |

```
|      4 | Divya    | 45000.00 |      20 |
|      5 | Esha     | 60000.00 |      30 |
|      6 | Farhan   | 35000.00 |      30 |
+--------+----------+----------+---------+
6 rows in set (0.00 sec)

+---------+-----------+----------+
| dept_id | dept_name | location |
+---------+-----------+----------+
|      10 | IT        | Delhi    |
|      20 | HR        | Mumbai   |
|      30 | Finance   | Pune     |
+---------+-----------+----------+
3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION; UPDATE employees SET salary = 85000.00 WHERE emp_id =
101; ALTER TABLE employee ADD COLUMN test_column INT; ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

ERROR 1146 (42S02): Table 'company_db.employees' doesn't exist
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

Query OK, 0 rows affected (0.00 sec)
```